

TaskXAI3

Interpretable models: Logistic regression, trees and RuleFit.

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1. Introducción

Imaginemos que nos han contratado para desplegar una serie de modelos predictivos para que los clientes pueden utilizarlos en sus negocios y entender cómo funcionan. El modo de desplegarlo será haciendo un repositorio público en Git, que estará disponible en GitHub.

Para facilitar la comprensión del modelo, utilizaremos Partial Dependence Plots, los cuales estarán comentados en el Report y estarán basados en lo que esté escrito por el R de aquí. Volvamos a la teoría y expliquemos cómo se pueden descomponer. La clasificación de los modelos según el tipo de interpretabilidad son: -Interpretable models: Regresores, Decision Trees -Black box models: Redes neuronales, SVM.. La clasificación de los modelos según el tipo de método de explicación son: -Model specific: solo sirve para un modelo concreto (como los coeficientes de regresión, que solo sirven en regresiones lineales). -Model-agnostic: La explicación sirve para CUALQUIER modelo, ya sea interpretable o de caja negra. Los métodos model-agnostic solo necesitan los Inputs y los Outputs para explicar el modelo, y encontramos técnicas como PDP, SHAP y LIME.

Vamos a aplicar una explicación model-agnostic PDP para explicar en 1D las bike rentals predichas por un Random Forest hecho con los bike rentals; Un PDP para explicar en 2D el número de bicicletas alquiladas según la Humidity y la temperatura; usar PDP en 1D para explicar el precio predicho con un random forest usando las features bedrooms, bathrooms, sqft_living, sqft_lot, floors y yr_built.

2. One-dimensional Partial Dependence Plot (Bike Rentals)

2.1 Objective

In this section, we analyze the marginal effect of individual features on the predicted number of bike rentals using a Random Forest model. The goal is to understand how each variable influences the model predictions independently.

2.2 Methodology

- Dataset: Bike rentals (day.csv)
- Target variable: cnt
- Model: Random Forest Regressor
- Features analyzed:
 - days_since_2011
 - temp
 - hum
 - windspeed

The model is trained using the preprocessed dataset described previously. Partial Dependence Plots (PDP) are computed for each selected feature to isolate their individual impact on predictions.

2.3 Data Processing & Modeling

```
library(randomForest)
```

```
## Warning: package 'randomForest' was built under R version 4.5.2
```

```
## randomForest 4.7-1.2
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
##  
## Adjuntando el paquete: 'ggplot2'
```

```
## The following object is masked from 'package:randomForest':  
##  
##     margin
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
##  
## Adjuntando el paquete: 'dplyr'
```

```
## The following object is masked from 'package:randomForest':  
##  
##      combine
```

```
## The following objects are masked from 'package:stats':  
##  
##      filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##      intersect, setdiff, setequal, union
```

```

set.seed(123)
#====DATA LOADING=====
rentals <- read.csv("day.csv")

#====DATA PROCESSING=====
# Selected variables
variables <- c("workingday", "holiday")

# One-hot encoding for season
rentals$season1 <- ifelse(rentals$season == 1, 1, 0)
rentals$season2 <- ifelse(rentals$season == 2, 1, 0)
rentals$season3 <- ifelse(rentals$season == 3, 1, 0)

variables <- append(variables, c("season1", "season2", "season3"))

# Weather variables
rentals$MISTY <- ifelse(rentals$weathersit == 2, 1, 0)
rentals$RAIN <- ifelse(rentals$weathersit %in% c(3, 4), 1, 0)

variables <- append(variables, c("MISTY", "RAIN"))

# Denormalize weather variables
rentals$temp <- rentals$temp * 41
rentals$hum <- rentals$hum * 100
rentals$windspeed <- rentals$windspeed * 67

variables <- append(variables, c("temp", "hum", "windspeed"))

# Days since 2011
rentals$dteday <- as.Date(rentals$dteday)
rentals$days_since_2011 <- as.integer(rentals$dteday - as.Date("2011-01-01"))

variables <- append(variables, "days_since_2011")

# Final modelling dataframe
rentals_model <- rentals[, c(variables, "cnt")]

#====MODELING: Random Forest model=====
rf_bikes <- randomForest(cnt ~ ., data = rentals_model, ntree = 500, importance = TRUE)

print(rf_bikes)

```

```

##
## Call:
## randomForest(formula = cnt ~ ., data = rentals_model, ntree = 500, importance = TRUE)
##
##           Type of random forest: regression
##           Number of trees: 500
## No. of variables tried at each split: 3
##
##           Mean of squared residuals: 431733.9
##           % Var explained: 88.48

```

```
importance(rf_bikes)
```

```
##              %IncMSE IncNodePurity
## workingday      8.992476      21458792
## holiday         2.497812       9825709
## season1        23.162746     265388062
## season2        13.224111     22873141
## season3        13.507912     69825591
## MISTY          20.166644     38445775
## RAIN           19.349701     59866535
## temp           44.803851     637112034
## hum            32.891520     184170312
## windspeed      19.399220     127912754
## days_since_2011 77.226854    1161002786
```

2.4 Visualizing 1-Dimensional PDP

We aren't not allowed to use the special library for pdp, **pdp**, so

```
library(pdp)
```

```
## Warning: package 'pdp' was built under R version 4.5.3
```

```

#----1D PDP for days since 2011
pdp_days <- partial(
  rf_bikes,
  pred.var = "days_since_2011",
  train = rentals_model,
  grid.resolution = 30
)

pdp_days = autoplot(pdp_days) +
  labs(
    title = "PDP: Days since 2011",
    x = "Days since 2011",
    y = "Predicted bike rentals"
  ) +
  theme_minimal()

#-----1D PDP for temperature
pdp_temp <- partial(
  rf_bikes,
  pred.var = "temp",
  train = rentals_model,
  grid.resolution = 30
)

pdp_temp = autoplot(pdp_temp) +
  labs(
    title = "PDP: Temperature",
    x = "Temperature (°C)",
    y = "Predicted bike rentals"
  ) +
  theme_minimal()

#-----1D PDP for humidity
pdp_hum <- partial(
  rf_bikes,
  pred.var = "hum",
  train = rentals_model,
  grid.resolution = 30
)

pdp_hum = autoplot(pdp_hum) +
  labs(
    title = "PDP: Humidity",
    x = "Humidity (%)",
    y = "Predicted bike rentals"
  ) +
  theme_minimal()

#-----1D PDP for wind speed
pdp_wind <- partial(
  rf_bikes,
  pred.var = "windspeed",
  train = rentals_model,
  grid.resolution = 30
)

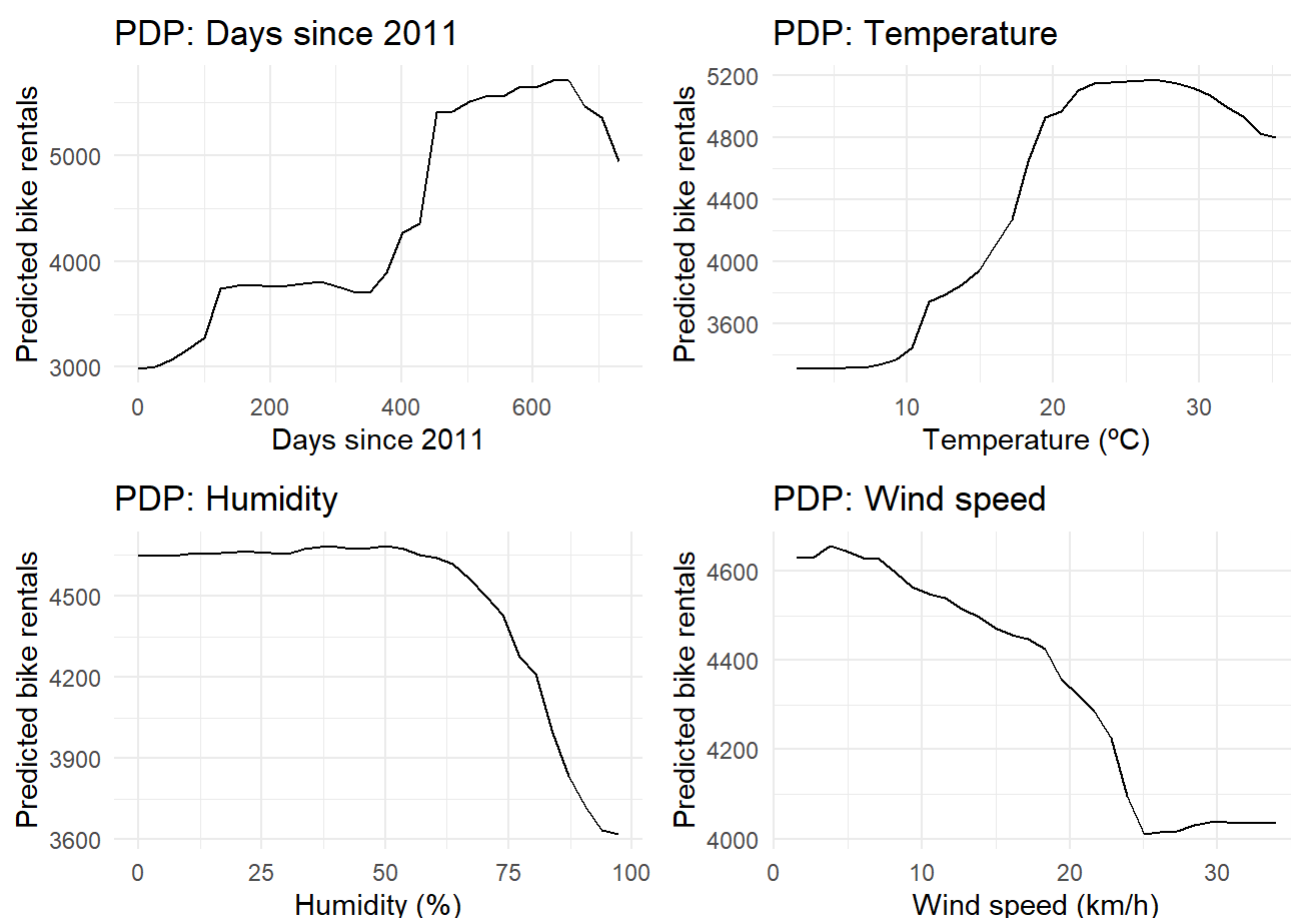
```

```
pdp_wind = autoplot(pdp_wind) +
  labs(
    title = "PDP: Wind speed",
    x = "Wind speed (km/h)",
    y = "Predicted bike rentals"
  ) +
  theme_minimal()
```

```
library(patchwork) #For showing the 4 graphs in a 2x2 matrix
```

```
## Warning: package 'patchwork' was built under R version 4.5.2
```

```
(pdp_days + pdp_temp) / (pdp_hum + pdp_wind)
```



2.5 Results and Interpretation

- **days_since_2011:** The model captures a clear upward trend over time, indicating increasing bike rental demand, followed by a slight stabilization or decline at later periods.
- **temp:** Temperature has a strong positive effect on rentals up to around 25°C, after which the demand slightly decreases, suggesting optimal weather conditions for biking.
- **hum:** Humidity shows a negative impact, especially at high levels (>70%), where rental demand drops significantly.
- **windspeed:** Wind speed negatively affects predictions; higher wind conditions lead to fewer expected rentals.

The PDP analysis reveals how each variable affects bike rental predictions while keeping other variables constant.

3. Bidimensional Partial Dependence Plot (Bike Rentals) for Temperature and Humidity

3.1 Objective

This section analyzes the joint effect of temperature and humidity on bike rental predictions using a 2D Partial Dependence Plot.

3.2 Methodology

- Variables analyzed: - temp - hum
- A random subset of the dataset is used to reduce computational cost.
- A 2D PDP is generated using a grid-based approach.
- The density distribution of both features is also visualized.

3.3 Data Preparing

The Random Forest model is already trained in the `rf_bikes` variable. However, a random subset of the dataset is used to reduce computational cost.

```
set.seed(123)

# Random sample to reduce computational cost
rentals_sample <- rentals_model %>%
  sample_n(min(500, nrow(rentals_model)))

head(rentals_sample)
```

	workingday	holiday	season1	season2	season3	MIS...	R...	temp	hum	
	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	
1	0	0	1	0	0	1	0	11.48000	51.5833	
2	0	0	0	1	0	0	0	17.93750	25.4167	
3	1	0	0	0	1	0	0	30.51085	63.4167	
4	0	0	0	1	0	0	0	29.14415	43.7083	
5	1	0	0	0	1	0	0	27.91415	47.6250	
6	1	0	0	1	0	1	0	25.31750	70.0833	

6 rows | 1-10 of 13 columns

3.4 Visualizing 2-Dimensional PDP

```

library(pdp)
library(ggplot2)

# Compute 2D PDP
pdp_temp_hum <- partial(
  rf_bikes,
  pred.var = c("temp", "hum"),
  train = rentals_sample,
  grid.resolution = 25
)

# 2D visualization
ggplot(pdp_temp_hum, aes(x = temp, y = hum, fill = yhat)) +

  # Heatmap tiles
  geom_tile(width = 1.8, height = 4) +

  # Density distribution of real observations
  geom_rug(
    data = rentals_sample,
    aes(x = temp, y = hum),
    inherit.aes = FALSE,
    alpha = 0.35
  ) +

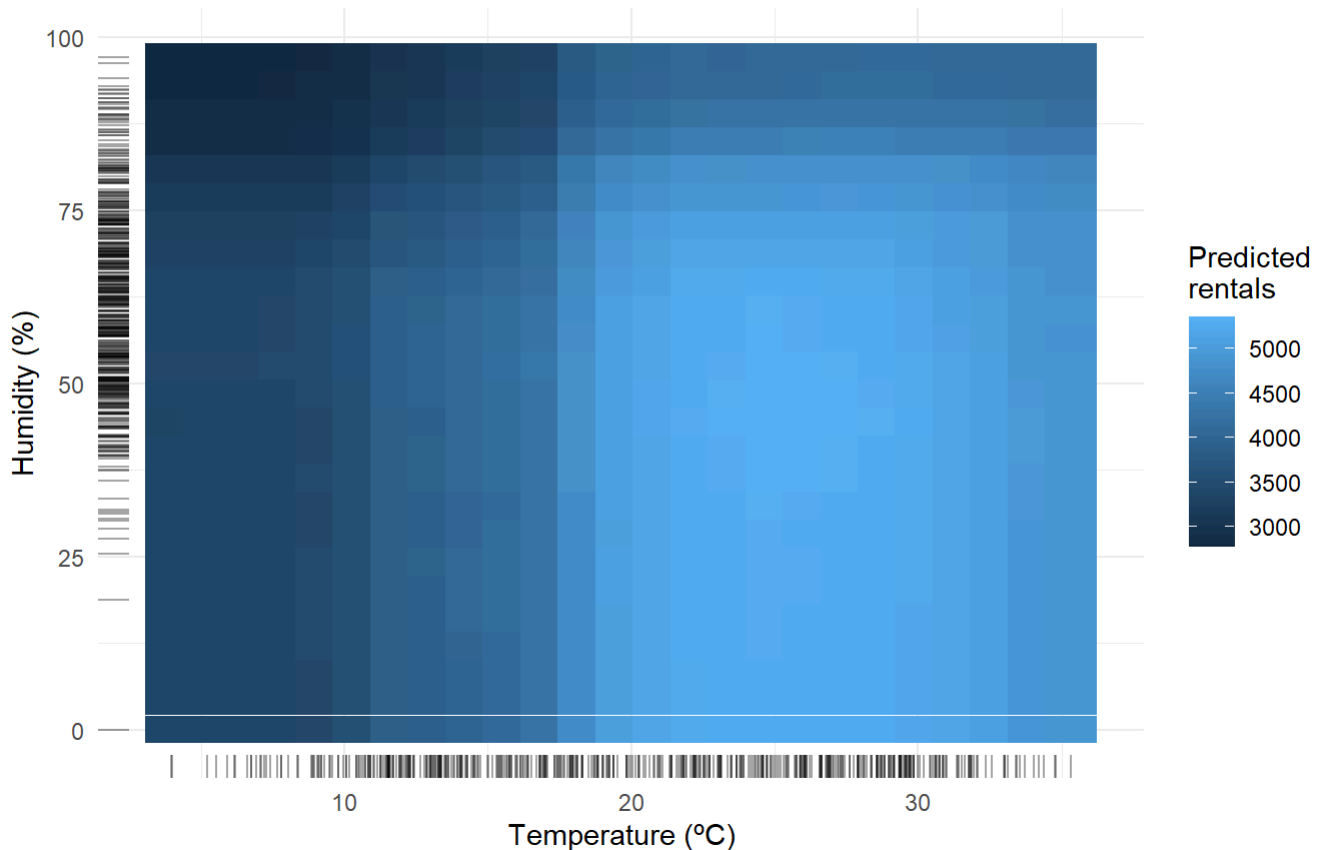
  labs(
    title = "2D PDP: Temperature and Humidity",
    subtitle = "Predicted bike rentals",
    x = "Temperature (°C)",
    y = "Humidity (%)",
    fill = "Predicted\nrentals"
  ) +

  theme_minimal()

```

2D PDP: Temperature and Humidity

Predicted bike rentals



3.5 Results and Interpretation

The 2D PDP allows identifying interaction effects between temperature and humidity: - **Low temperature & high humidity:** the model predicts the lowest number of bike rentals under cold and humid weather conditions. - **High temperature & low humidity:** the model predicts the highest rental demand when temperatures are warm, especially between 20°C and 30°C, and humidity remains moderate. - **Interaction effect:** temperature has stronger impact on predictions than humidity. Rental demand increases significantly with temperature, while very high humidity slightly reduces the positive effect of warm weather.

4. PDP for House Price Prediction

4.1 Objective

To analyze the effect of structural house features on predicted house prices using a Random Forest model.

4.2 Methodology

Dataset: `kc_house_data.csv`

Target variable: - price

Features: - bedrooms - bathrooms - sqft_living - sqft_lot - floors - yr_built

A random subset of the dataset is used to optimize computation time.

4.3 Data Processing & Modeling

The dataset is loaded and prepared for modelling. A Random Forest model is trained using the selected structural house features.

```
library(randomForest)
library(dplyr)

#####Load dataset
houses <- read.csv("kc_house_data.csv")

# Selected features
house_variables <- c(
  "bedrooms",
  "bathrooms",
  "sqft_living",
  "sqft_lot",
  "floors",
  "yr_built"
)

# Final dataframe
houses_model <- houses[, c(house_variables, "price")]

# Removing missing values
houses_model <- na.omit(houses_model)

#####Random sample to reduce computational cost
set.seed(123)

houses_sample <- houses_model %>%
  sample_n(min(3000, nrow(houses_model)))

#####Train Random Forest model
rf_houses <- randomForest(
  price ~ .,
  data = houses_sample,
  ntree = 500,
  importance = TRUE
)

print(rf_houses)
```

```
##
## Call:
## randomForest(formula = price ~ ., data = houses_sample, ntree = 500,      importance = TR
UE)
##              Type of random forest: regression
##              Number of trees: 500
## No. of variables tried at each split: 2
##
##              Mean of squared residuals: 58059020598
##              % Var explained: 56.54
```

```
importance(rf_houses)
```

```
##          %IncMSE  IncNodePurity
## bedrooms    12.81854  2.158059e+13
## bathrooms   23.22041  7.322781e+13
## sqft_living  59.51594  1.692828e+14
## sqft_lot     19.87702  4.605898e+13
## floors       17.55243  1.411023e+13
## yr_built     42.82163  4.406349e+13
```

4.4 Visualizing 1-Dimensional PDP

The following PDPs show the marginal effect of each selected variable on predicted house price.

```

# Bedrooms
pdp_bedrooms <- partial(
  rf_houses,
  pred.var = "bedrooms",
  train = houses_sample,
  grid.resolution = 20
)

pdp_bedrooms = autoplot(pdp_bedrooms) +
  labs(
    title = "PDP: Bedrooms",
    x = "Bedrooms",
    y = "Predicted price"
  ) +
  theme_minimal()

# Bathrooms
pdp_bathrooms <- partial(
  rf_houses,
  pred.var = "bathrooms",
  train = houses_sample,
  grid.resolution = 20
)

pdp_bathrooms = autoplot(pdp_bathrooms) +
  labs(
    title = "PDP: Bathrooms",
    x = "Bathrooms",
    y = "Predicted price"
  ) +
  theme_minimal()

# sqft_living
pdp_sqft <- partial(
  rf_houses,
  pred.var = "sqft_living",
  train = houses_sample,
  grid.resolution = 30
)

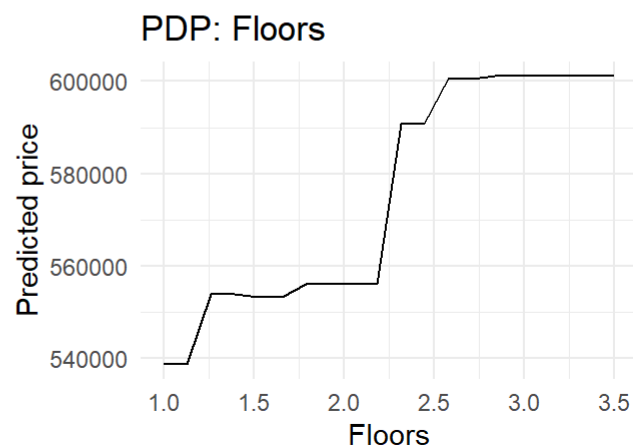
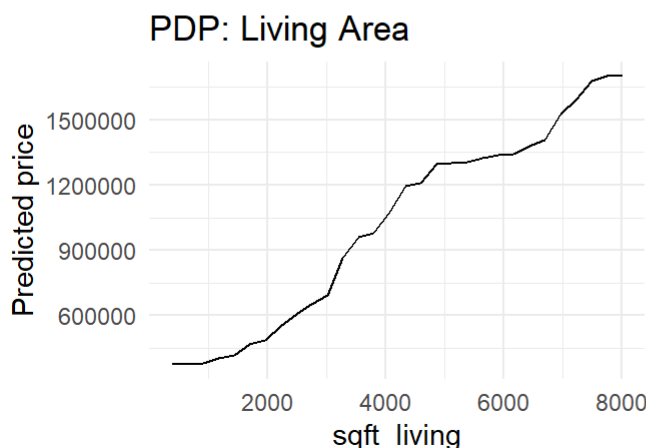
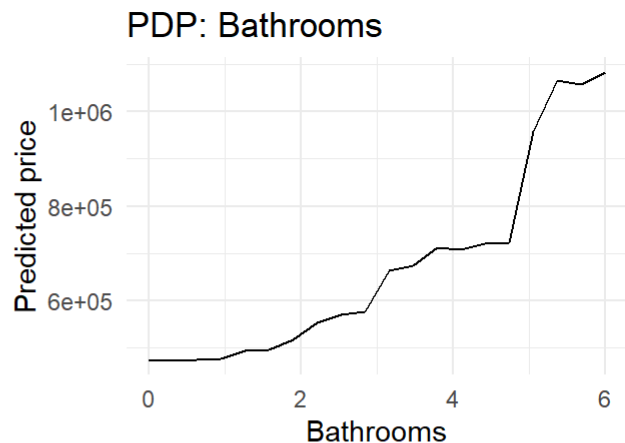
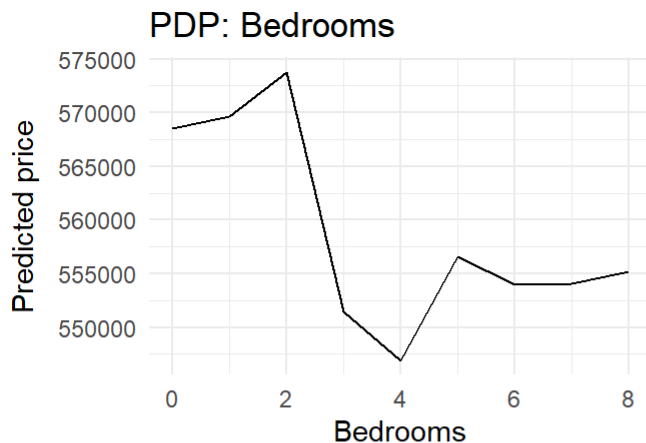
pdp_sqft = autoplot(pdp_sqft) +
  labs(
    title = "PDP: Living Area",
    x = "sqft_living",
    y = "Predicted price"
  ) +
  theme_minimal()

# Floors
pdp_floors <- partial(
  rf_houses,
  pred.var = "floors",
  train = houses_sample,
  grid.resolution = 20
)

```

```
pdp_floors = autoplot(pdp_floors) +
  labs(
    title = "PDP: Floors",
    x = "Floors",
    y = "Predicted price"
  ) +
  theme_minimal()

(pdp_bedrooms + pdp_bathrooms) / (pdp_sqft + pdp_floors)
```



4.5 Results and Interpretation

- **bedrooms:** the relationship is not strictly linear. Houses with 2 bedrooms achieve the highest predicted prices, while larger numbers of bedrooms tend to reduce or stabilize the prediction price.
- **bathrooms:** bathrooms show a strong positive effect on house prices. Properties with more bathrooms are consistently associated with higher predicted values.
- **sqft_living:** living area is one of the most influential variables. Predicted price increases steadily as the living area grows.
- **floors:** increasing the number of floors moderately increases predicted prices. The curve seems like a stair because there are no houses with 1.5 floors, only works in entire numbers.

5. Conclusions

In the bike rental scenario, the analysis revealed that temperature is the most influential environmental factor, while high humidity and strong wind conditions negatively affect rental demand. The 2D PDP also showed that the interaction between temperature and humidity plays an important role in prediction behavior.

In the housing price scenario, the living area and number of bathrooms were identified as the most relevant variables for price estimation. The PDPs also revealed non-linear relationships for variables such as bedrooms and floors.

Overall, PDPs provide valuable insights into how predictive models behave while maintaining model flexibility and high predictive performance. These techniques improve transparency and support business decision-making.